Supplementary Materials

Anonymous submission

Preliminaries

Pre-LN Transformer Block

A core module commonly used in an FM is the Pre-LN Transformer block shown in Figure S1. Let $X_\ell \in \mathbb{R}^{s\times d}$ denote the hidden representation at the ℓ-th Transformer layer, where s is the sequence length and d is the feature dimension. For simplicity, we assume all layers share the same hidden dimension. A standard Pre-LN Transformer block consists of a multi-head self-attention sub-layer followed by a gated feed-forward network (FFN), each with residual connections:

$$X'_\ell = X_\ell + \mathrm{Attn}(\mathrm{LN}(X_\ell)),$$
$$X_{\ell+1} = X'_\ell + \mathrm{FFN}(\mathrm{LN}(X'_\ell)).$$

Here, $\mathrm{LN}(\cdot)$ denotes layer normalization. Following the LLaMA architecture, the FFN adopts a gated form:

$$\mathrm{FFN}(X) = (\sigma(XW_\mathrm{g}) \odot XW_\mathrm{u})\, W_\mathrm{d},$$

where $\sigma(\cdot)$ is the activation function and $\odot$ denotes element-wise multiplication. The multi-head self-attention module is defined as

$$\mathrm{Attn}(X) = \mathrm{Concat}\left(\mathrm{head}_1(X), \ldots, \mathrm{head}_h(X)\right) W_O,$$

where each attention head is computed by

$$\mathrm{head}_i(X) = \mathrm{softmax}\left(\frac{XW_Q^i (XW_K^i)^\top}{\sqrt{d_\mathrm{head}}}\right) XW_V^i.$$

Here, h is the number of attention heads, and d_head is the head dimension. The matrices $W_Q^i, W_K^i, W_V^i \in \mathbb{R}^{d\times d_\mathrm{head}}$ denote the query, key, and value projections of the i-th head, respectively. In addition, $W_O \in \mathbb{R}^{d\times d}$ is the output projection, while $W_\mathrm{g}, W_\mathrm{u} \in \mathbb{R}^{d\times d_\mathrm{ffn}}$ and $W_\mathrm{d} \in \mathbb{R}^{d_\mathrm{ffn}\times d}$ are the gate, up, and down projections in the gated FFN.

Theoretical Analysis of SDI

Basic Definitions

This section develops a theoretical analysis of SDI through its application to the shifted SiLU gate at position P_5 of a SwiGLU block. We follow the notation of the main paper. For a projection module p in layer ℓ, let $z \sim \mu_{\ell,p}$ be its local hidden-state input and let $\phi(z)$ be the corresponding local feature map. Because we study one module, we omit (ℓ, p) and write

$$z \sim \mu, \qquad z \in \mathbb{R}^d, \qquad \phi(z) \in \mathbb{R}^n. \tag{S1}$$

Here d is the local input dimension and n is the number of features. At the MLP gate position P_5, z is the normalized hidden state entering the gate and up projections. For the local analysis, we assume that z is distributed symmetrically about the origin: z and $-z$ have the same distribution. This implies $\mathbb{E}[z] = 0$. We further assume that z has nonsingular covariance and finite fourth moment:

$$\Sigma := \mathrm{Cov}(z) = \mathbb{E}[zz^\top] \succ 0, \quad \mathbb{E}\|z\|_2^4 < \infty. \tag{S2}$$

The original SFLI summarizes feature diversity through the Gram matrix

$$M(\phi, \mu) = \mathbb{E}_{z\sim\mu}[\phi(z)\phi(z)^\top], \tag{S3}$$

and defines its ϵ-rank as

$$r_\epsilon(M) = \#\{j : \lambda_j(M) > \epsilon\}. \tag{S4}$$

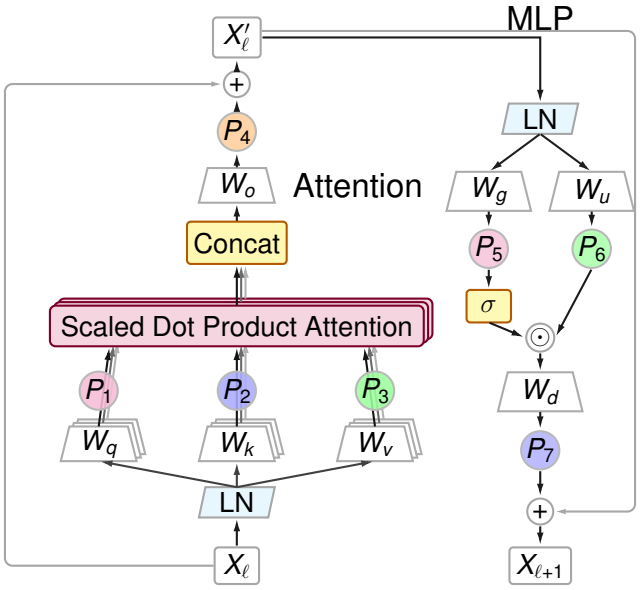


Figure S1: Pre-LN Transformer block with candidate SDI insertion positions P_1–P_7.

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For SDI, the matrix introduced below measures a different quantity, termed *local feature accessibility*, which characterizes the part of the feature map that is linearly accessible from the local input.

In the main paper, SDI initializes a nonlinear response as $\sigma(\gamma(Az+c))$. Although SDI is applied to the gate projection at P_5, the feature available to the subsequent down projection is the complete SwiGLU product. With $\sigma = \mathrm{SiLU}$, we therefore take

$$\phi(z) := \mathrm{SiLU}(\gamma(Az + c)) \odot W_u z \in \mathbb{R}^n. \tag{S5}$$

Here $A, W_u \in \mathbb{R}^{n\times d}$, $c \in \mathbb{R}^n$, and $\gamma > 0$. SiLU is applied elementwise, and $\odot$ denotes elementwise multiplication.

Definition S1 (Input–feature coupling matrix)**.** For the fixed local distribution μ, define

$$T(\phi) = \frac{1}{\sqrt{n}}\mathbb{E}[\phi(z)z^\top]\Sigma^{-1/2}. \tag{S6}$$

Here $\Sigma^{-1/2}$ denotes the symmetric inverse square root of Σ. When ϕ is clear from context, we write $T = T(\phi)$. This matrix measures how strongly the features $\phi(z)$ are related to the input z. It is easy to check that T is linear in its feature-map argument:

$$T(\alpha f + \beta g) = \alpha T(f) + \beta T(g) \tag{S7}$$

for feature maps f and g with the same output dimension. For $\epsilon > 0$, its local ϵ-rank is

$$r_\epsilon(T) = \#\{j : s_j(T) > \epsilon\}, \tag{S8}$$

where $s_j(T)$ are the singular values of T.

Example S2. Let $z = (z_1, z_2, z_3)^\top \sim \mathcal{N}(0, I_3)$ and consider the feature map

$$\phi(z) = (z_1, z_2, z_1 + z_2)^\top. \tag{S9}$$

Since $\Sigma = I_3$, the input–feature matrix is

$$T = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}. \tag{S10}$$

Its singular values are 1, $1/\sqrt{3}$, and 0. Therefore, $r_\epsilon(T) = 2$ for every $0 < \epsilon < 1/\sqrt{3}$. The feature map loses all input information in the z_3 direction. Thus, the local ϵ-rank based on T provides an intuitive measure of how many input directions contribute to the output neurons.

Why SDI Improves Local Feature Accessibility

Let

$$\phi_0(z) := \mathrm{SiLU}(\gamma Az) \odot W_u z, \tag{S11}$$

be the zero-shift feature map and define $T_0 := T(\phi_0)$. The following result shows how the shift c changes T in the small-γ regime.

Theorem S3 (First-order effect of a gate shift)**.** *Under the assumptions on z stated above, as $\gamma \to 0$, with μ, A, W_u, and c fixed,*

$$T_0 = O(\gamma^2), \tag{S12}$$

$$T = \frac{\gamma}{2\sqrt{n}}\,\mathrm{diag}(c)W_u\Sigma^{1/2} + O(\gamma^2). \tag{S13}$$

The $O(\gamma^2)$ terms are measured in spectral norm, with constants independent of γ.

Proof. Using $\mathrm{SiLU}(t) = t/(1 + e^{-t})$, we have

$$\mathrm{SiLU}(t) = \frac{t}{2}\left(1 + \tanh\frac{t}{2}\right). \tag{S14}$$

Since $|\tanh x| \le |x|$,

$$\left|\mathrm{SiLU}(t) - \frac{t}{2}\right| \le \frac{t^2}{4}. \tag{S15}$$

Let $a_i^\top$ and $u_i^\top$ denote the ith rows of A and W_u, respectively, and let $q_i(z) = a_i^\top z + c_i$. The global bound (S15) gives

$$\begin{aligned} \phi_i(z) &= \frac{\gamma}{2}q_i(z)(u_i^\top z) + R_i(z), \\ |R_i(z)| &\le \frac{\gamma^2}{4}|q_i(z)|^2|u_i^\top z|. \end{aligned} \tag{S16}$$

Let $R(z) = (R_1(z), \dots, R_n(z))^\top$. To bound $T(R)$, use $|u_i^\top z| \le \|u_i\|_2\|z\|_2$ and $|q_i(z)| \le \|a_i\|_2\|z\|_2 + |c_i|$. Since a_i, u_i, and c_i are fixed, there is a constant K_i, independent of γ, such that

$$|R_i(z)|\|z\|_2 \le K_i\gamma^2\big(1 + \|z\|_2^4\big). \tag{S17}$$

For each i, we have

$$\begin{aligned} \|T(R_i)\|_2 &\le \|\Sigma^{-1/2}\|_2\mathbb{E}[|R_i(z)|\|z\|_2] \\ &\le K_i\gamma^2\|\Sigma^{-1/2}\|_2\mathbb{E}\big[1 + \|z\|_2^4\big] =: L_i\gamma^2. \end{aligned} \tag{S18}$$

By (S2), L_i is finite and independent of γ. The full matrix $T(R)$ is

$$T(R) = \frac{1}{\sqrt{n}}\begin{pmatrix} T(R_1) \\ \vdots \\ T(R_n) \end{pmatrix}. \tag{S19}$$

Therefore,

$$\|T(R)\|_2 \le \|T(R)\|_F \le \left(\frac{1}{n}\sum_{i=1}^n L_i^2\right)^{1/2}\gamma^2 =: L\gamma^2. \tag{S20}$$

Let R_0 be the remainder obtained by setting $c = 0$. By the linearity of T, the coordinate expansion above gives

$$T_0 = \frac{\gamma}{2}T\big((Az) \odot (W_u z)\big) + T(R_0), \tag{S21}$$

$$T = \frac{\gamma}{2}T\big((Az) \odot (W_u z)\big) + \frac{\gamma}{2}T\big(c \odot (W_u z)\big) + T(R). \tag{S22}$$

For each i, the symmetry assumption gives

$$\mathbb{E}[(a_i^\top z)(u_i^\top z)z^\top]\Sigma^{-1/2} = 0, \tag{S23}$$

because the integrand is odd under $z \mapsto -z$. Therefore,

$$T\big((Az) \odot (W_u z)\big) = 0. \tag{S24}$$

The shift term is

$$T\big(c \odot (W_u z)\big) = \frac{1}{\sqrt{n}}\,\mathrm{diag}(c)W_u\Sigma^{1/2}. \tag{S25}$$

Thus,

$$T_0 = T(R_0), \tag{S26}$$

$$T = \frac{\gamma}{2\sqrt{n}} \operatorname{diag}(c) W_u \Sigma^{1/2} + T(R). \tag{S27}$$

The same remainder estimate with $c = 0$ gives a constant L_0, independent of γ, such that

$$\|T(R_0)\|_2 \le L_0 \gamma^2. \tag{S28}$$

This completes the proof. □

For small γ, the shift c provides the leading contribution to T. If $\operatorname{diag}(c) W_u \Sigma^{1/2}$ has full column rank, then T has d singular values of order γ. The following corollary expresses this separation in terms of the local ϵ-rank.

Corollary S4 (Rank separation)**.** *Under the conditions of Theorem S3, suppose $n \ge d$. Let $B = \operatorname{diag}(c) W_u \Sigma^{1/2}$, suppose B has full column rank, and let $s_d(B) > 0$ be its smallest singular value. There exist $K > 0$ and $\gamma_0 > 0$ such that, for every $0 < \gamma < \gamma_0$, there exists $\epsilon > 0$ satisfying*

$$K\gamma^2 < \epsilon < \frac{\gamma}{2\sqrt{n}} s_d(B) - K\gamma^2 \tag{S29}$$

for which

$$r_\epsilon(T_0) = 0, \qquad r_\epsilon(T) = d. \tag{S30}$$

Proof. By Theorem S3, there exist $K > 0$ and $\gamma_1 > 0$ such that, for every $0 < \gamma < \gamma_1$,

$$\|T_0\|_2 \le K\gamma^2, \qquad \left\| T - \frac{\gamma}{2\sqrt{n}} B \right\|_2 \le K\gamma^2. \tag{S31}$$

Choose $0 < \gamma_0 \le \gamma_1$ such that $2K\gamma_0 < s_d(B)/(2\sqrt{n})$. For every $0 < \gamma < \gamma_0$, this gives

$$K\gamma^2 < \frac{\gamma}{2\sqrt{n}} s_d(B) - K\gamma^2. \tag{S32}$$

Thus, there exists $\epsilon > 0$ satisfying

$$K\gamma^2 < \epsilon < \frac{\gamma}{2\sqrt{n}} s_d(B) - K\gamma^2. \tag{S33}$$

Since $s_1(T_0) = \|T_0\|_2$ is the largest singular value of T_0, for every j,

$$s_j(T_0) \le s_1(T_0) = \|T_0\|_2 \le K\gamma^2 < \epsilon \tag{S34}$$

and hence $r_\epsilon(T_0) = 0$.

For T, the singular-value perturbation bound gives

$$s_d(T) \ge s_d\left(\frac{\gamma}{2\sqrt{n}} B\right) - \left\| T - \frac{\gamma}{2\sqrt{n}} B \right\|_2 \tag{S35}$$

$$\ge \frac{\gamma}{2\sqrt{n}} s_d(B) - K\gamma^2 > \epsilon. \tag{S36}$$

Thus $s_j(T) \ge s_d(T) > \epsilon$ for $j = 1, \dots, d$, and therefore $r_\epsilon(T) = d$. □

Thus, for sufficiently small γ, the shift introduced by SDI raises the local ϵ-rank from 0 to d at a suitable threshold ϵ.

The local ϵ-rank measures the input directions captured by the features, but not how much the MLP changes the residual stream. To study this, let $x \sim \nu$, $x \in \mathbb{R}^d$, be the residual-stream vector before the MLP branch, and set $z = \mathcal{N}(x)$, whose distribution is μ. The residual block is

$$F(x) = x + W_d \phi(z), \tag{S37}$$

where $\mathcal{N}$ is the normalization layer and $W_d \in \mathbb{R}^{d \times n}$ is the down projection. Expectations involving x below are taken with respect to ν.

Corollary S5 (Stability of the Jacobian)**.** *Assume that $\mathcal{N}$ is differentiable ν-almost everywhere and*

$$\mathbb{E}\big[(1 + \|z\|_2)^2 \|J_{\mathcal{N}}(x)\|_2^2\big] < \infty. \tag{S38}$$

For fixed A, W_u, c, and W_d, there exists a constant $K > 0$ such that

$$\mathbb{E}\|J_F(x) - I\|_F^2 \le K\gamma^2 \tag{S39}$$

Proof. Write $\sigma = \mathrm{SiLU}$. Differentiating ϕ_i gives

$$\begin{aligned} \nabla_z \phi_i(z) = \gamma \sigma'\big(\gamma(a_i^\top z + c_i)\big)\, a_i (u_i^\top z) \\ + \sigma\big(\gamma(a_i^\top z + c_i)\big)\, u_i. \end{aligned} \tag{S40}$$

Let $L_\sigma = \sup_t |\sigma'(t)|$. Since $|\sigma(t)| \le |t|$, the norm of row i is bounded by

$$\begin{aligned} \|\nabla_z \phi_i(z)\|_2 \le \gamma L_\sigma \|a_i\|_2 \|u_i\|_2 \|z\|_2 \\ + \gamma\big(\|a_i\|_2 \|z\|_2 + |c_i|\big) \|u_i\|_2. \end{aligned} \tag{S41}$$

Summing the squared row bounds gives a finite constant K_ϕ, depending only on the fixed feature parameters, for which

$$\|J_\phi(z)\|_F \le K_\phi \gamma (1 + \|z\|_2). \tag{S42}$$

Applying the chain rule to (S37) gives, for ν-almost every x,

$$J_F(x) - I = W_d J_\phi(z) J_{\mathcal{N}}(x). \tag{S43}$$

Therefore,

$$\begin{aligned} \|J_F(x) - I\|_F &\le \|W_d\|_2 \|J_\phi(z)\|_F \|J_{\mathcal{N}}(x)\|_2 \\ &\le \|W_d\|_2 K_\phi \gamma (1 + \|z\|_2) \|J_{\mathcal{N}}(x)\|_2. \end{aligned} \tag{S44}$$

Taking expectations and using (S38) yields

$$\mathbb{E}\|J_F(x) - I\|_F^2 \le K\gamma^2, \tag{S45}$$

where

$$K := \|W_d\|_2^2 K_\phi^2 \mathbb{E}\big[(1 + \|z\|_2)^2 \|J_{\mathcal{N}}(x)\|_2^2\big]. \tag{S46}$$

□

Together with Corollary S4, this result separates the two effects of SDI: the shift raises the local ϵ-rank, while the scale controls the change to the residual block.

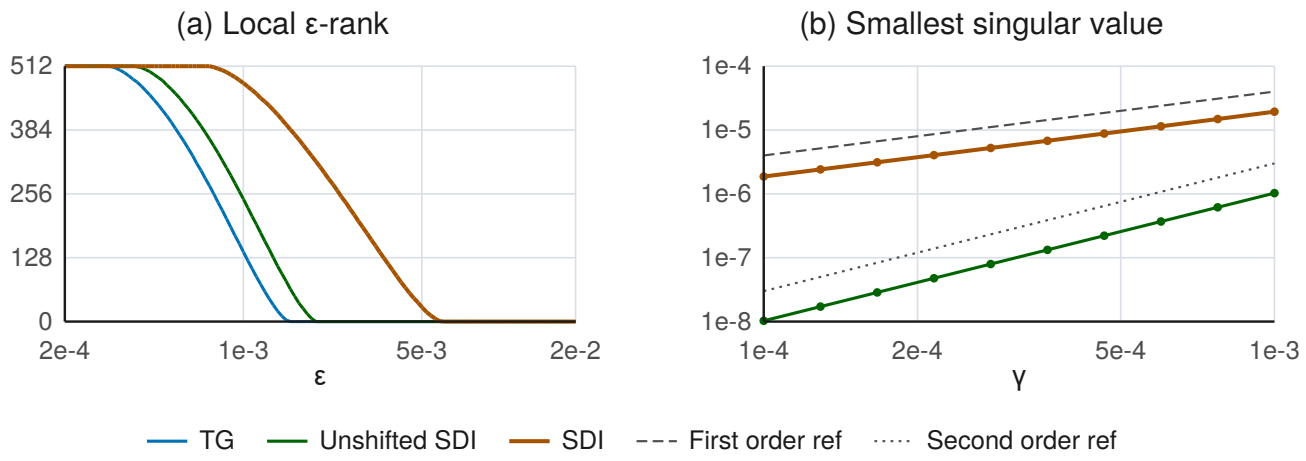


Figure S2: Local rank for the SwiGLU block. (a) The local ϵ-rank $r_\epsilon(T)$ at the same nominal gate scale. (b) The smallest singular value $s_d(T)$ as γ varies, with reference lines of slopes 1 and 2. Curves are computed from ten shared seeds.

Examples

This section provides an intuitive illustration of the effect of SDI.

Example S6 (SwiGLU block in LLaMA-71M)**.** Consider a SwiGLU block with $d = 512$ and $n = 1368$. We compare the following three gate initializations:

$$\begin{aligned} \text{Default:} \quad & W_g \sim \text{TruncNormal}(0, \sigma_g^2), \quad b_g = 0, \\ \text{Unshifted SDI:} \quad & W_g = \gamma A, \quad b_g = 0, \\ \text{SDI:} \quad & W_g = \gamma A, \quad b_g = \gamma c. \end{aligned}$$

Here A and c follow the SDI initialization in the main paper, and γ controls the gate scale. Default is the standard initialization commonly used in the large language models, while Unshifted SDI uses the same A as SDI but sets the shift to zero. The three methods use the same batch of $1,536$ inputs and the same W_u. Each test is repeated over ten seeds.

We conduct two tests: (a) fix $\sigma_g = \gamma = 0.02$ and vary ϵ; (b) vary γ.

Figure S2(a) shows the local ϵ-rank as the threshold ϵ varies. For small ϵ, all three matrices have full rank. As ϵ increases, the ranks of Default and Unshifted SDI decrease earlier, while SDI retains a substantially higher rank.

At $\epsilon = 10^{-3}$, the median ranks are

Initialization	$r_\epsilon(T)$
Default	140
Unshifted SDI	246
SDI	478

At the same gate scale, SDI therefore captures substantially more input directions above the selected threshold than Default and Unshifted SDI.

Figure S2(b) explains the rank advantage observed in Figure S2(a). Without the shift, $s_d(T)$ scales quadratically with γ, while with the shift it scales linearly, consistent with Theorem S3. The shift therefore maintains larger singular values at small gate scales, supporting the higher local ϵ-rank of SDI.

Supplementary Background Knowledge

Weight Initialization in Deep Neural Networks

The choice of parameter initialization plays a critical role in the training dynamics of deep neural networks. In the absence of a proper initialization scheme, activations and gradients can either vanish or explode as they propagate through layers, leading to unstable optimization dynamics and poor convergence. Over the past decade, a substantial body of research has sought to identify principled initialization methods that maintain stable signal variance across layers. Two particularly influential approaches are the *Xavier initialization* (Glorot and Bengio 2010) and the *Kaiming initialization* (He et al. 2015), each designed to address variance preservation under specific activation functions.

Motivation: Preserving Signal Variance Consider a feed-forward neural network composed of L layers, where each layer computes

$$h^{(l)} = \phi(W^{(l)} h^{(l-1)} + b^{(l)}),$$

with nonlinearity $\phi(\cdot)$ and weight matrix $W^{(l)} \in \mathbb{R}^{n_l \times n_{l-1}}$. If the elements of $W^{(l)}$ are drawn from a zero-mean distribution with variance $\text{Var}[W_{ij}^{(l)}] = \sigma^2$, then the variance of activations satisfies

$$\text{Var}[h^{(l)}] = \text{Var}[\phi(W^{(l)} h^{(l-1)})].$$

To ensure stable learning, one seeks to maintain $\text{Var}[h^{(l)}] \approx \text{Var}[h^{(l-1)}]$ and likewise for backpropagated gradients. This "variance preservation" criterion prevents exponential growth or decay of signals through depth, and thus guides the derivation of modern initialization schemes.

Xavier (Glorot) Initialization Glorot and Bengio (2010) derived an initialization scheme that balances the variance of activations and gradients for networks employing symmetric activation functions such as $\tanh$ or the logistic sigmoid. By analyzing deep networks with sigmoid activations, they

observed that inappropriate initialization of the weights W drives pre-activations into the saturated regions of the neuron $x \mapsto \sigma(Wx + b)$, leading to nearly constant activations and vanishing gradients. Under mild independence assumptions between weights and inputs, and requiring unit variance of pre-activations, they propose to set

$$\mathrm{Var}\big[W_{ij}^{(l)}\big] = \frac{2}{n_{l-1} + n_l},$$

where n_{l-1} and n_l denote the fan-in and fan-out of layer l. This corresponds to sampling weights either from a uniform distribution

$$W_{ij}^{(l)} \sim \mathcal{U}\left(-\sqrt{\frac{6}{n_{l-1} + n_l}}, \sqrt{\frac{6}{n_{l-1} + n_l}}\right),$$

or from a zero-mean normal distribution with variance $\mathrm{Var}\big[W_{ij}^{(l)}\big] = \frac{2}{n_{l-1}+n_l}$. This initialization, often referred to as *Xavier* or *Glorot* initialization (the uniform variant is commonly called *Glorot uniform*), helps maintain approximately constant variance for both forward activations and backpropagated gradients across layers, and empirically stabilizes the training of deep networks with sigmoid or hyperbolic tangent nonlinearities by mitigating the vanishing gradient problem.

Kaiming (He) Initialization Subsequent work by He et al. (2015) extended this analysis to rectified linear unit (ReLU) activations, which zero out roughly half of their inputs and thereby reduce signal variance by approximately a factor of $1/2$. To compensate, they proposed scaling the initialization variance to

$$\mathrm{Var}[W_{ij}^{(l)}] = \frac{2}{n_{l-1}},$$

or equivalently sampling

$$W_{ij}^{(l)} \sim \mathcal{N}\left(0, \frac{2}{n_{l-1}}\right).$$

This choice, commonly referred to as *Kaiming* or *He initialization*, preserves the expected variance of activations under ReLU-like nonlinearities, maintaining stable forward and backward signal propagation. Empirical results demonstrated that this initialization substantially improves convergence speed and final accuracy in very deep convolutional networks, such as VGG and ResNet architectures.

Both Xavier and Kaiming initializations can be viewed as instances of the general principle of *variance scaling initialization*, in which the variance of weights is inversely proportional to the fan-in or fan-out of each layer. These schemes form the foundation of most contemporary initialization practices and are implemented by default in modern deep learning frameworks. While originally derived under simplifying assumptions of independence and Gaussianity, they continue to serve as effective heuristics even in complex architectures, highlighting the robustness of variance-preserving initialization in deep learning.

Truncated Gaussian Initialization with Small Variance Small-variance, zero-mean Gaussian initialization is a standard choice in modern deep and Transformer-based architectures (Vaswani et al. 2017). It has been used since the early days of neural networks due to its simplicity and intuitive appeal. A typical instantiation is $W \sim \mathcal{N}(0, 0.02^2)$, which is commonly used in models such as BERT (Devlin et al. 2019), Mistral (Jiang et al. 2023) and Qwen3 (Yang et al. 2025). Moreover, the initialization can be modified by applying truncation to the sampled values. Specifically, samples are restricted to the interval $[-2\sigma, 2\sigma]$, and any values outside this range are rejected and redrawn. This approach is commonly known as truncated Gaussian initialization.

Structured First-Layer Initialization By mirroring the spectral-method principle that linearly independent basis functions enhance approximation power, as well as the observed staircase phenomenon in which sharp drops in training loss coincide with increases in the ϵ-rank of neuron functions, Tang et al. (2025) proposed *Structured First-Layer Initialization* (SFLI). SFLI aims to improve the initial feature diversity by constructing approximately ϵ-linearly independent neuron functions in the first hidden layer. Given

$$F_i(\mathbf{x}) = \sigma(\mathbf{w}_i^\top \mathbf{x} + b_i),$$

SFLI parameterizes the initial weights and biases as

$$\mathbf{w}_i = \gamma \boldsymbol{\alpha}_i, \qquad b_i = \gamma c_i,$$

such that

$$F_i(\mathbf{x}) = \sigma\big(\gamma(\boldsymbol{\alpha}_i^\top \mathbf{x} + c_i)\big).$$

Here, the shifts c_i are distributed across the input domain to separate the neuron functions, while $\boldsymbol{\alpha}_i$ determines their orientations. The localization scale is recommended as

$$\gamma = C\frac{n^{1/d} - 1}{|\Omega|^{1/d}}, \qquad C \in [0.5, 2],$$

where n is the layer width, d is the input dimension, and Ω is the input domain. The construction is adapted to the activation function: Gaussian directions are used for cosine activation, unit-sphere directions for hyperbolic tangent and hat activations, and spatially distributed centers for Gaussian radial features. SFLI modifies only the first-layer initialization and leaves the subsequent architecture and optimization procedure unchanged. Experiments on function approximation and PDE benchmarks show that SFLI increases the initial ϵ-rank, accelerates convergence, mitigates spectral bias, and improves prediction accuracy with negligible computational overhead.

Detailed Configurations of Main Experiments

LLaMA Experiments

We provide the training configurations and hyperparameters used for the LLaMA pretraining experiments reported in Table 1 of the main paper. Unless otherwise specified, the settings follow the corresponding baseline configurations. Table S1 summarizes the main hyperparameters across model scales.

All LLaMA models are trained with the Adam optimizer and a cosine learning-rate schedule. We use an initial learning rate of 1×10^{-3} for models up to LLaMA-250M, and 5×10^{-4} for LLaMA-1B. The learning rate is linearly warmed up during the first 10% of training and then decayed according

Params	Hidden	Intermediate	Heads	Layers	Steps	Data Amount	LR	Total Batch Size
71M	512	1368	8	12	10K	1.1B	1×10^{-3}	512
130M	768	2048	12	12	20K	2.2B	1×10^{-3}	512
250M	1024	2560	16	24	40K	3.9B	1×10^{-3}	512
1B	2048	5461	32	24	100K	5.0B	5×10^{-4}	512

Table S1: Hyperparameters for the LLaMA main experiments.

Params	Hidden	Intermediate	Heads	Layers	Steps	LR	Total Batch Size
200M	512	1376	8	16	40K	1×10^{-3}	512
300M	640	1728	10	25	50K	1×10^{-3}	512
0.5B	768	4096	12	24	100K	5×10^{-4}	512
6B	3584	18944	28	21	13K	5×10^{-4}	512

Table S2: Hyperparameters for the Qwen main experiments.

to a cosine schedule to 10% of its initial value. Across all experiments, the maximum sequence length is set to 256, and the total batch size is set to 512.

Qwen Experiments

For the Qwen experiments reported in Table 2 of the main paper, we use training configurations following the same principle as the LLaMA experiments. All models are trained with the Adam optimizer and a cosine learning-rate schedule. We use an initial learning rate of 1×10^{-3} for models smaller than 0.5B parameters, and 5×10^{-4} for models with 0.5B parameters or more. The learning rate is linearly warmed up during the first 10% of training and then decayed by a cosine schedule to 10% of its initial value. Across all Qwen experiments, the maximum sequence length is set to 256, and the total batch size is set to 512. Other hyperparameters used in Qwen experiments are presented in Table S2.

Vision Transformer Experiments

For the Vision Transformer (ViT) experiments reported in Table 3 of the main paper, we use the training settings summarized in Table S3. The evaluated models are DeiT-Tiny/16 and DeiT-Small/16, where DeiT denotes Data-efficient Image Transformer. DeiT largely follows the standard ViT architecture, while improving the training recipe through stronger data augmentation, regularization, and optional distillation. In our experiments, both models are trained on ImageNet-1K at an input resolution of 224^2 for 120 epochs, using AdamW with a cosine learning-rate schedule, linear warmup, mixup, and cutmix.

Transolver Experiments

For the Transolver experiments reported in the main paper, we use the training settings summarized in Table S4. We evaluate Transolver on three 2D structured-mesh datasets: Airfoil, Pipe, and Plasticity. All experiments use a hidden dimension of 128, eight attention heads, eight layers, 64 slices, a learning rate of 1×10^{-3}, and gradient clipping at 0.1. The MLP ratio and batch size are adjusted according to the dataset, as detailed in the table.

training configurations	**DeiT-Tiny/16 ImageNet-1K** 224^2	**DeiT-Small/16 ImageNet-1K** 224^2
optimizer	AdamW	AdamW
base lr	5×10^{-4}	5×10^{-4}
weight decay	0.05	0.05
batch size	256	256
training epochs	120	120
lr schedule	cosine decay	cosine decay
warmup epochs	5	5
warmup schedule	linear	linear
mixup	0.8	0.8
cutmix	1.0	1.0

Table S3: Training settings for Vision Transformer experiments.

WaveFormer Experiments

For the WaveFormer experiments reported in Table 3 of the main paper, we use the training settings summarized in Table S5. We evaluate WaveFormer-Tiny and WaveFormer-Small on ImageNet-1K at an input resolution of 224^2. Both models use a patch size of 4 and are trained for 120 epochs with AdamW, a cosine learning-rate schedule, five warmup epochs, and a batch size of 128. We apply mixup and cutmix to both models, while setting the drop-path rates to 0.1 and 0.2 for WaveFormer-Tiny and WaveFormer-Small, respectively.

Supplementary Experiments

In the LLaMA experiments section of the main paper, we report that a set of experiments was conducted to discover the best activation function for SDI. To select an activation for attention-side SDI, we apply SDI only to the query projection in all decoder layers of LLaMA-130M and vary the activation while holding all other hyperparameters fixed. We evaluate $\tanh$, cosine, sinc, and SiLU. Table S6 shows that cosine-SDI performs best in this preliminary comparison, motivating its

Exp.	Hidden	Heads	Layers	Slices	MLP Ratio	Batch Size	LR	Grad Clip
Airfoil	128	8	8	64	1	4	1×10^{-3}	0.1
Pipe	128	8	8	64	2	8	1×10^{-3}	0.1
Plasticity	128	8	8	64	1	8	1×10^{-3}	0.1

Table S4: Training settings for Transolver experiments.

training configurations	**WaveFormer-Tiny ImageNet-1K** 224^2	**WaveFormer-Small ImageNet-1K** 224^2
patch size	4	4
optimizer	AdamW	AdamW
base lr	1×10^{-3}	1×10^{-3}
minimum lr	1×10^{-6}	1×10^{-6}
weight decay	0.05	0.05
batch size	128	128
training epochs	120	120
lr schedule	cosine decay	cosine decay
warmup epochs	5	5
drop path rate	0.1	0.2
MLP ratio	4.0	4.0
post norm	True	True
layer scale	None	None
mixup	0.8	0.8
cutmix	1.0	1.0

Table S5: Training settings for WaveFormer experiments.

Method	Eval Loss (↓)	Eval PPL (↓)
q-only with tanh-SDI	3.26437	26.16
q-only with cos-SDI	**3.24269**	**25.60**
q-only with sinc-SDI	3.27433	26.43
q-only with SiLU-SDI	3.24396	25.64

Table S6: Preliminary comparison of activation-specific SDI variants for attention-side insertion. We apply SDI only to the query projection in all decoder layers of LLaMA-130M, while keeping the training setup fixed. This motivates the use of cos-SDI for attention-side position ablation.

use in the subsequent attention-side position ablation. We present the results of this set of experiments here. As shown in Table S6, cos-SDI achieves the best performance in this preliminary comparison, which motivates the adoption of this method for attention-side ablation.

Parameter Overhead

Table S7 reports the additional trainable parameters introduced by SDI in the LLM experiments. We report both the number of added parameters and their percentages relative to the corresponding base model sizes, for LLaMA and Qwen families. Across all evaluated models, these additional parameters account for less than 0.03% of the corresponding base model size, demonstrating that SDI incurs negligible parameter overhead.

Family	**Model**	**Added Parameters**	**Relative Overhead**
LLaMA	LLaMA-71M	0.02M	0.028%
	LLaMA-130M	0.02M	0.015%
	LLaMA-250M	0.06M	0.024%
	LLaMA-1B	0.13M	0.013%
Qwen	Qwen-200M	0.02M	0.010%
	Qwen-300M	0.04M	0.013%
	Qwen-0.5B	0.10M	0.020%
	Qwen-6B	0.40M	0.007%

Table S7: Additional parameters introduced by SDI in the evaluated LLMs. Relative overhead is computed with respect to the corresponding base model size.

Experimental Compute Resources

The experiments were conducted on a server with the following configuration:

- 2 Intel Xeon Platinum 8358P CPUs (64 physical cores / 128 threads in total)
- 1.9 TiB of system memory
- 8 NVIDIA A100-SXM4-80GB GPUs
- Ubuntu 22.04.4 LTS operating system

In addition, the main software libraries and frameworks used in the experiments include:

- Python 3.9.25
- PyTorch 2.8.0
- Transformers 4.49.0
- Datasets 4.5.0
- PEFT 0.17.1
- BitsAndBytes 0.48.2
- CUDA 12.x Python libraries
- W&B 0.25.1

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